

NELSON DANG

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EDUCATION

Douglas College

Diploma in Computer Science

New Westminster, BC

Sep. 2023 – Sep. 2025

Simon Fraser University

Bachelor's in Computer Science

Surrey, BC

Jan. 2026 – Expected April 2029

EXPERIENCE

Software Developer

May 2026 – Present

Tochi Desserts LTD.

Vancouver, BC

- Cut weekly bookkeeping 70% (5 hrs → 1.5 hrs) by designing and deploying a full-stack Next.js/PostgreSQL inventory platform used by 10+ staff, replacing manual spreadsheet tracking
- Designed a real-time finance dashboard tracking revenue, cost of goods, gross profit, and margins with automated COGS calculation from inventory stock usage
- Implemented role-based access control with three permission levels (admin, manager, staff) using Supabase Auth and PostgreSQL row-level security policies
- Integrated live inventory tracking with low-stock alerts, stock adjustment history, and automatic cost calculations tied to item usage

Manager

May 2024 – Present

Tochi Desserts LTD.

Vancouver, BC

- Ran full truck operations in the owner's absence, leading 8 staff through peak service and reconciling 700+ transactions on the busiest day of the FIFA Fan Festival
- Coordinated staff scheduling, event bookings, and client and event-organizer relationships, ensuring coverage for daily service and off-site catering
- Directed staffing, supply, and transport logistics for 6–10 events monthly — including multi-day festivals (FIFA Fan Festival, PNE, Christmas markets) — sustaining 100+ consecutive events with zero ingredient stockouts

PROJECTS

Mental Health Chatbot | *Python, Flask, SQLite, Google Gemini API, JavaScript*

Jun 2026 – Jul 2026

- Built a full-stack mental health support chatbot that classifies user messages into 20+ emotional states using VADER sentiment analysis, keyword scoring, and spaCy noun-phrase extraction
- Engineered an LLM integration layer over the Gemini REST API, injecting rolling conversation history and classified emotional state into structured prompts, with a deterministic guardrail routing crisis-flagged messages outside the model entirely
- Designed a crisis-detection safety net using word-boundary phrase matching with an idiom-exception list to reduce false positives, routing flagged messages to a hardcoded crisis-hotline response independent of the AI model
- Built a Flask/SQLAlchemy backend persisting per-session conversation history, plus Web Speech API voice input/output

Escape Prison Break Game | *Java, Maven, JUnit*

Jan 2026 – April 2026

- Built a 4-level 2D grid-based prison escape game in Java with a Swing GUI, featuring pixel-art sprite rendering, animated walk cycles, and a player-centered scrolling camera viewport
- Implemented a breadth-first search pathfinding algorithm driving enemy AI state transitions between patrol and pursuit, triggered by a 5-tile line-of-sight check across a 4-level grid
- Applied the Factory Method design pattern to decouple all level configuration (spawn coordinates, coin requirements, exit positions) from the game engine, enabling clean addition of new levels
- Wrote 22 JUnit test suites achieving about 80% coverage across the game engine, board, score manager, entity classes, and all four level factories

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, HTML/CSS

Frameworks: React, Next.js, Node.js, Flask, SQLAlchemy, Tailwind CSS

Developer Tools: Git, REST APIs, IntelliJ IDEA, VS Code, Visual Studio, Vercel, Supabase, JUnit, Maven

Interests: Weight lifting, Video games, Running, Pickleball